

Aaron Celestian, Ph.D.

CURATOR OF MINERAL SCIENCES
NATURAL HISTORY MUSEUM OF LOS ANGELES COUNTY

Mineral scientist bridging fundamental research with real-world applications in astrobiology, energy materials, and environmental science. Recognized expert in ion exchange mechanisms and biosignature detection. Research spans from searching for life on Mars to developing sustainable lithium extraction methods, with a focus on how minerals interact with their environments and living systems to address global challenges.

EDUCATION

Ph.D. in Geosciences

Stony Brook University, 2006

Dissertation: Time-resolved structural characterization of microporous silicotitanates and niobium silicotitanates

M.S. in Geosciences

Stony Brook University, 2002

Thesis: Ion exchange of $K_8Al_8Ge_8O_{32} \cdot 8H_2O$ and $Na_{24}Al_{24}Ge_{24}O_{96} \cdot 40H_2O$ gismondine

B.S. in Geology with Emphasis in Mineralogy

University of Arizona, 1999

PROFESSIONAL EXPERIENCE

CURRENT POSITIONS

Curator, Mineral Sciences

2021 – Present

Natural History Museum of Los Angeles County

Adjunct Associate Professor

2016 – Present

Department of Earth Sciences, University of Southern California

Adjunct Instructor

2021 – Present

West Los Angeles College

PREVIOUS POSITIONS

Science and Research Affiliate NASA Jet Propulsion Laboratory	2018 – 2025
Associate Curator, Mineral Sciences Natural History Museum of Los Angeles County	2016 – 2021
Associate Professor (Tenured) Department of Geography and Geology, Western Kentucky University	2012 – 2015
Director, Advanced Materials Institute Western Kentucky University	2010 – 2015
Assistant Professor Department of Geography and Geology, Western Kentucky University	2007 – 2012
Visiting Assistant Professor School of Earth and Environmental Science, City University of New York Queens College	2006 – 2007

RESEARCH INTERESTS

Astrobiology — Biosignatures in minerals, life detection on Mars and icy moons

Energy Materials — Sustainable lithium extraction, ion exchange materials

Environmental Science — Photocatalytic degradation, metal sequestration in zeolites

PharmaMineralogy — Kidney stone formation, bacterial-mineral interactions

Cultural Heritage — Gemstone provenance analysis, art conservation

FUNDING & GRANTS

NOAA Sea Grant (2023–2025)

Plastics Past and Present: Understanding historical trends of microplastic consumption in California marine food webs · Co-PI with W. Ludt

Department of Energy Critical Materials (2021–2023)

Informed design of crystalline ion exchangers: Improved λ -MnO₂ phase for lithium extraction from geologic fluids · Co-PI with C. Camiré, W. Bourcier

American Elements (2021–2022)

Selective Rare Earth Element Ion Exchange in Minerals

Getty Museum (2020–2022)

PST 2024 — Realities of the Environmental Disparities that People of Color Face · Co-PI with M. Garcia, B. H. Jaramillo, M. Muro, L. Higgins

NASA RT&D (2020/2021–2023)

Simulating Ocean World Microbiology and Pressure-Relevant Ecological Systems · Co-PI with S. Perl

Oak Ridge National Laboratory (2021)

Neutron Scattering Studies of Microporous Zirconium Silicates

NASA NSPIRES (2018–2021)

Investigating a novel role for iron redox cycling in the lithification of microbial mats · Co-PI with F. Corsetti, V. Petryshyn, S. S. Bradley, J. R. Spear, W. Berelson

NSF EPSCoR (2013–2014)

Advancing porous nanominerals for basic energy science, environmental security, and materials genomics

American Chemical Society — Petroleum Research Fund (2011–2013)

Mechanisms of ion selectivity in porous frameworks

PUBLICATIONS

172 peer-reviewed articles · [View on Google Scholar \(https://scholar.google.com/citations?user=Yp2uw-0AAAAJ&hl=en\)](https://scholar.google.com/citations?user=Yp2uw-0AAAAJ&hl=en)

2026 – 7 PUBLICATIONS

Mechanism of Lithium Ion Exchange Into Spinel Manganese Oxide

Celestian, Aaron J, Bourcier, William, Camiré, Christopher L, Kumar, Jayanthi, Paranthaman, Mariappan Parans

Journal of Raman Spectroscopy, 57(1), 131–139

Intercalated bacterial biofilms are intrinsic internal components of calcium-based kidney stones

Schmidt, William C, Mousavi, Ava, Li, Jiahui, Yang, Rena, et al.

Proceedings of the National Academy of Sciences, 123(5), e2517066123

Mineral Weathering Experiments for Terrestrial Planetary Analogs in a Continuous-Flow Reactor

Weber, Jessica M, Martinez, Eduardo, Sheppard, Rachel Y, Rodriguez, Laura E, Celestian, Aaron, et al.

Geofluids, 2026(1), 6180315

Molecular Mechanisms of Magnesium Sulfate Crystallization: Bond Length Inversion and the Role of Hydration in Mineral Formation

Celestian, Aaron J, Xu, Wenqian, Kubicki, James D, Perl, Scott M

American Mineralogist

Nitrogen mineral hosts in subduction zone metamorphosed ultramafic rocks

Cooperdock, Emily HG, Bonilla, Ailani N, Grabiec, Justine G, Celestian, Aaron J, et al.

Chemical Geology, 123307

Perchlorate Anaerobiosis in *Haloferax volcanii*: A Novel Metabolic Capability for A Model Haloarchaeon

Robinson, Adam, McQuaig-Ulrich, Shannon, Dondero, Tyler, Celestian, Aaron, Perl, Scott

Cadvanite, a New Cadmium Vanadate from the Burro Mine, San Miguel County, Colorado, USA

Kampf, Anthony R, Celestian, Aaron J, Ma, Chi, Rose, Timothy P, Marty, Joe

The Canadian Journal of Mineralogy and Petrology, 64(2), 185–193

2025 – 12 PUBLICATIONS

Evaporitic Preservation of Modern Carotenoid Biomarkers and Halophilic Microorganisms in Mars Analog Hypersaline Environments

Perl, Scott M, Baxter, Bonnie K, Celestian, Aaron J, et al.

Astrobiology, 25(11), 765–776

Evidence for Low Dissolved Silica in mid-Mesozoic Oceans

Yager, Joyce A, West, A Joshua, Trower, Elizabeth J, Fischer, Woodward W, Celestian, Aaron J, et al.

American Journal of Science, 325, 1

Svornostite-(NH₄): a new mineral from the Blue Lizard mine, San Juan County, Utah, USA, and the establishment of the svornostite mineral group

Kampf, Anthony R, Olds, Travis A, Plášil, Jakub, Ma, Chi, Celestian, Aaron J, Marty, Joe
Mineralogical Magazine, 89(5), 609–617

MP21-12 Biofilm-Forming Bacteria Found Within Calcium-Based Kidney Stones

Mousavi, Ava, Schmidt, William C, Li, Jiahui, Yang, Rena, Celestian, Aaron, Scotland, Kymora B
The Journal of Urology, 213(5S), e709

Exploring Mechanisms of Microbialite Texture Formation through mapping in situ CaCO₃ precipitation and Microbial Cell Position

Cassady, Victoria C, Petryshyn, Victoria A, Celestian, Aaron J, et al.
Goldschmidt 2025 Conference

RamanLab: A Comprehensive Cross-Platform Software Suite for Advanced Raman Spectroscopy Analysis

Celestian, Aaron J
Authorea Preprints

Mapping quartz and feldspar areal abundance and grain sizes using integrated VTIR data

Grant, Kathleen, Brodrick, Philip Gregory, Celestian, Aaron, West, A Joshua, et al.
Authorea Preprints

Machine Learning Strategies to Distinguish Iron Sulfide, Oxide, and (Oxy)hydroxide Minerals Using Raman and LIBS Spectroscopy

Lamm, SN, Sheppard, RY, Celestian, AJ, Barge, LM, et al.
LPI Contributions, 3090, 2702

Evidence for Acidic Bleaching and Dominant Groundwater Processes in Endmember Hypersaline and Acidic Lakes in Western Australia

Plattner, TA, Schmidt, BE, Celestian, A, et al.
LPI Contributions, 3090, 2811

Whole rock and mineral separate nitrogen concentration and isotopic data for meta-ultramafic rocks

Cooperdock, Emily, Bonilla, Ailani, Grabiec, Justine, Celestian, Aaron, et al.
Interdisciplinary Earth Data Alliance (IEDA) dataset, 237

2024 — 19 PUBLICATIONS

Signature Specimen: Elbaite from the Cruzeiro Pegmatite, Minas Gerais, Brazil

Francis, Carl A, Celestian, Aaron
Rocks & Minerals, 99(1), 86–88

Radiocarbon Chronology/Growth Rates of Ooids from Great Salt Lake, Utah

Paradis, Olivia P, Corsetti, Frank A, Celestian, Aaron, et al.
Utah Geological Association Publication

In situ potassium and hydrogen ion exchange into a cubic zirconium silicate microporous material

Lively, Jason, Celestian, Aaron J
PLOS ONE, 19(3), e0298661

MP63-13 Bacteria Trigger Calcium Oxalate Crystal Formation

Nazarian, Reyhaneh, Lin, Neil, Celestian, Aaron, Scotland, Kymora B, et al.
Journal of Urology, 211(5S), e1036

PD63-06 Exploring Bacterial Roles in Calcium Oxalate Stone Formation Through High-Resolution Imaging

Mousavi, Ava, Nazarian, Reyhaneh, Schmidt, William C, Celestian, Aaron, Scotland, Kymora B, et al.
Journal of Urology, 211(5S), e1294

The Biology of Biosignature Detection — Year One: Building the Experimental Framework for Extant and Extinct Life Measurements

Perl, Scott M, Cockell, Charles S, Fischer, Woodward W, Celestian, Aaron, et al.
2024 Astrobiology Science Conference

Increasing equitable access to meaningful research experiences through undergraduate mineralogy and earth materials classes

Cooperdock, Emily HG, Grabiec, Justine G, Rojas, Alexia, Celestian, Aaron J
2024 Goldschmidt Conference

The Mineralogy of Laboratory Simulated Hydrothermal Vents as Analogs to Saponite-Rich Hydrothermal Environments on Early Earth and Mars

Carman, Nancy, Hausrath, Elisabeth, Celestian, Aaron J, et al.
AbSciCon 2024, 421-084

Fe/Mg-silicate chemical gardens as analogs to silicate-rich hydrothermal chimneys on early Earth and Mars

Carman, Nancy A, Hausrath, Elisabeth M, Celestian, Aaron, et al.
ACS Earth and Space Chemistry, 8(10), 1982–1996

Living to Lithified: Construction and Preservation of Silicified Biomarkers

Rasmussen, Kalen L, Celestian, Aaron J, Corsetti, Frank A, et al.
Geobiology, 22(5), e12620

Insight Into Mineral Fabrics, Biofabrics, and Channel Architecture to Comprehensively Characterize Microbialite Texture

Cassady, Victoria, Petryshyn, Victoria A, Celestian, Aaron, et al.
Geological Society of America Abstracts, 56, 405114

Tracking Composition of Searles Lake Brines and Mineral Fluid Inclusions with Progressive Evaporation via Raman and LIBS

Nguyen, Melonie, Celestian, Aaron, Perl, Scott M, Corsetti, Frank
Geological Society of America Abstracts, 56, 404422

Beyond the Stones: the Transformative Impact of Mineral Collections at the Natural History Museum of Los Angeles County

Leftwich, Kristin, Kampf, Anthony R, Celestian, Aaron

A Tale of Two Seasons: The Geochemical Evolution of Acidic Hypersaline Lakes in Western Australia

Plattner, Taylor, Schmidt, Britney, Celestian, Aaron, et al.

AGU Fall Meeting Abstracts, 2024, EP52A-06

Unveiling Hosts for Nitrogen Transportation and Storage in Meta-Serpentinites

Bonilla, Ailani, Grabiec, Justine G, Cooperdock, Emily HG, Celestian, Aaron, et al.

AGU Fall Meeting Abstracts, 2024

Nitrogen in Ultramafic and Sedimentary Mélange Matrices and its Implications for Subduction Zone Nitrogen Budgets

Grabiec, Justine, Bonilla, Ailani, Cooperdock, Emily HG, Celestian, Aaron J, et al.

Geological Society of America Abstracts, 56, 403278

The Biology of Biosignature Detection: Rationale and Experimental Frameworks for Biological Validation

Perl, Scott, Cockell, Charles, Fischer, Woodward, Celestian, Aaron, et al.

Characterization of mineral fabrics, biofabrics, and channel architecture in incipient microbialites

Cassady, Victoria, Petryshyn, Victoria A, Celestian, Aaron, et al.

AGU Fall Meeting Abstracts, 2024, B53J-08

The Biology of Biosignature Detection: Framework, Experiments, and Science Mission Concepts

Perl, SM, Cockell, CS, Fischer, WW, Celestian, AJ, et al.

LPI Contributions, 3007, 3526

Rare Earth: Crystalline Treasures — Exhibition at the Santa Barbara Museum of Natural History

Celestian, Aaron J

Rocks & Minerals, 98(2), 148–159

New Mineral Names: Arsenic and Lead

Celestian, Aaron J

American Mineralogist, 108(3), 634–636

Micro-elemental Analysis of Stromatolites Reveal Fe as Potential Biosignature and Remnant of a Lithification Mechanism

Cassady, Victoria C, Levene, Dylan R, Celestian, Aaron J, Petryshyn, Victoria A, et al.

Goldschmidt 2023 Conference

Microbial Pigment Robustness in Martian and Ocean Worlds Brines and Evaporites

Perl, SM, Celestian, AJ, Cockell, CS, Corsetti, FA, et al.

54th Lunar and Planetary Science Conference, 2806, 2434

Detection and Preservation of Extremophile Biosignatures in Lithium-Bearing Evaporites for Life Detection

Javier, BM, Perl, SM, Basu, C, Fischer, W, Celestian, AJ

54th Lunar and Planetary Science Conference, 2806, 2825

New Mineral Names: Heavy Metal and Minerals from China

Celestian, Aaron J

American Mineralogist, 108(10), 1993–1994

Rate-Dependent Preservation of Psychrotolerant Halophiles within Cryobrine Systems

Perl, SM, Cockell, CS, Celestian, AJ, Javier, BJ, Basu, C

Brines Across the Solar System, 2689, 2041

Biosignature Preservation Potential of Li-Containing Hypersaline Environments

Javier, BM, Perl, SM, Basu, C, Fischer, W, Celestian, AJ

Examining Fe-Cycling in Microbial Mats as a Lithification Mechanism

Petryshyn, Victoria, Berelson, Will, Spear, John, Celestian, Aaron, et al.

Geological Society of America Abstracts, 55, 393626

Micro-Elemental Analysis of Stromatolites Reveal Fe as Potential Biosignature

Rasmussen, Kalen, Levene, Dylan R, Celestian, Aaron J, Petryshyn, Victoria A, et al.

Thirty-third Annual Goldschmidt Conference, 20694

2022 – 14 PUBLICATIONS

Assessing the origin of pisoids within a travertine system in the border of Puna Plateau, Argentina

Mors, R Agustin, Gomez, Fernando J, Celestian, Aaron J, Corsetti, Frank A, et al.

Sedimentology, 69(3), 1252–1275

Relianceite-(K), a new phosphate–oxalate mineral from the Rowley mine, Arizona, USA

Kampf, Anthony R, Cooper, Mark A, Celestian, Aaron J, Ma, Chi, Marty, Joe

Mineralogical Magazine, 86(4), 539–547

Dendoraite-(NH₄), a new phosphate–oxalate mineral from the Rowley mine, Arizona, USA

Kampf, Anthony R, Cooper, Mark A, Celestian, Aaron J, Ma, Chi, Marty, Joe

Mineralogical Magazine, 86(4), 531–538

New Mineral Names: Alteration Products

Celestian, Aaron J

American Mineralogist, 107(2), 318–320

Rapid phase identification of apatite and zircon grains for geochronology using X-ray micro-computed tomography

Cooperdock, Emily HG, Hofmann, Florian, Celestian, Aaron J, et al.

Geochronology, 4(2), 501–515

New Mineral Names: High-Pressure and Precious Minerals

Celestian, Aaron J

American Mineralogist, 107(4), 778–780

Flaggite, a new mineral with stair-step-like HCP layers from Tombstone, Arizona, USA

Kampf, Anthony R, Mills, Stuart J, Celestian, Aaron J, Ma, Chi, et al.

Mineralogical Magazine, 86(3), 397–404

Diagenesis and Subsurface Habitable Environments I Online Session

Sheppard, Rachel, Perl, Scott M, Celestian, Aaron J, et al.

2022 Astrobiology Science Conference

Evidence for benthic oxygen production in Neoproterozoic lacustrine stromatolites

Wilmeth, Dylan T, Lalonde, Stefan V, Celestian, Aaron J, Corsetti, Frank A, et al.

Geology, 50(8), 907–911

Geobiological impacts of Earth's early complex bioturbators during the Ediacaran-Cambrian transition

Cribb, Alison T, Celestian, Aaron J, Berelson, William M, Corsetti, Frank A, et al.

2022 Goldschmidt Conference

New Mineral Names: Hydrous Minerals

Celestian, Aaron J

American Mineralogist, 107(8), 1656–1658

Rapid phase identification of apatite and zircon grains for geochronology using micro-computed tomography

Cooperdock, Emily HG, Hofmann, Florian, Celestian, Aaron J, et al.

2022 Goldschmidt Conference

Biogenic stromatolites as lacustrine oxygen oases ~2.7 Ga

Wilmeth, Dylan T, Lalonde, Stefan V, Petryshyn, Victoria A, Celestian, Aaron J, et al.
2022 Goldschmidt Conference

Evidence for Rapid Rates of Oxygenic Photosynthesis Within Neoproterozoic Stromatolites (abstract)

Wilmeth, Dylan T, Corsetti, Frank A, Berelson, William M, Celestian, Aaron J, et al.
Geological Society of America Abstracts, 50, 318717

2021 – 12 PUBLICATIONS

Jasonsmithite, a new phosphate mineral with a complex microporous framework, from the Foote mine, North Carolina, USA

Kampf, Anthony R, Celestian, Aaron J, Nash, Barbara P
American Mineralogist, 106(2), 174–179

Thebaite-(NH₄), a new phosphate–oxalate mineral from the Rowley mine, Arizona, USA

Kampf, Anthony R, Cooper, Mark A, Celestian, Aaron J, Nash, Barbara P, Marty, Joe
Mineralogical Magazine, 85(3), 379–386

Plastic ingestion by freshwater turtles: a review and call to action

Clause, Adam G, Celestian, Aaron J, Pauly, Gregory B
Scientific Reports, 11(1), 5672

Allantoin and natrosulfatourea, two new bat-guano minerals from the Rowley mine, Arizona, USA

Kampf, Anthony R, Celestian, Aaron J, Nash, Barbara P, Marty, Joe
The Canadian Mineralogist, 59(3), 603–616

A proposed geobiology-driven nomenclature for astrobiological in situ observations and sample analyses

Perl, Scott M, Celestian, Aaron J, Cockell, Charles S, Corsetti, Frank A, et al.
Astrobiology, 21(8), 954–967

New Mineral Names: Diamonds, Dumps, and Fumaroles

Celestian, Aaron J

American Mineralogist, 106(12), 2028–2030

Mineralogy and History of Semi-Precious Gemstones on the Borghese-Windsor Cabinet

Celestian, Aaron, Heginbotham, Arlen, Ehlmann, Bethany, et al.

Geological Society of America Abstracts, 53, 368743

Salty Environments: The importance of evaporites and brine environments as habitats and preservers of biosignatures

Perl, Scott, Celestian, Aaron J, Cockell, Charles S, Corsetti, Frank A, et al.

Bulletin of the American Astronomical Society, 53(4), 240

Classification of Iron (oxy)hydroxides and Sulfides Using Mission-Ready Spectroscopic Techniques and Machine Learning

Lamm, Sarah, Rodriguez, Laura E, Perl, Scott M, Celestian, Aaron J, Barge, Laura M

Geological Society of America Abstracts, 53, 368733

Preservation of Dynamic Biological Processes from Extant Halophilic Life

Perl, SM, Celestian, AJ, Cockell, CS, Basu, C, et al.

Brines in the Solar System: Modern Brines, 2614, 6048

Comparison of Microporous Minerals for Potential Contaminant Uptake

Osorio, Marvin, Oze, Christopher, Celestian, Aaron

Geochemistry of Chromium-Silicate Minerals

Dall, Jacqueline, Oze, Christopher, Celestian, Aaron, Rossman, George

2020 – 3 PUBLICATIONS

Examining the role of Fe-cycling in microbial mat lithification and stromatolite abundance through Earth history

Corsetti, Frank A, Berelson, William, Celestian, Aaron J, et al.

AGU Fall Meeting Abstracts, 2020, PP013-01

New emerald locality in Southern California

Celestian, Aaron J

Colored Stones and Organic Materials

Celestian, Aaron J, Edley, Richard

Gems & Gemology, 56(4)

2019 – 17 PUBLICATIONS

Neoarchean (2.7 Ga) lacustrine stromatolite deposits in the Hartbeesfontein Basin, South Africa

Wilmeth, DT, Corsetti, FA, Beukes, NJ, Celestian, AJ, et al.

Precambrian Research, 320, 291–302

In situ Cs and H exchange into gaidonnayite and proposed mechanisms of ion diffusion

Celestian, Aaron J, Lively, Jason, Xu, Wenqian

Inorganic Chemistry, 58(3), 1919–1928

Aragonite dissolution kinetics and calcite/aragonite ratios in sinking and suspended particles in the North Pacific

Dong, Sijia, Berelson, William M, Celestian, Aaron J, et al.

Earth and Planetary Science Letters, 515, 1–12

Camanchacaite, chinchorroite, espadaite, and three additional new minerals from the Torrecillas mine, Chile

Kampf, Anthony R, Nash, Barbara P, Celestian, Aaron J, Dini, Maurizio, Molina Donoso, Arturo A

Mineralogical Magazine, 83(5), 655–671

Phoxite, the first phosphate-oxalate mineral

Kampf, Anthony R, Celestian, Aaron J, Nash, Barbara P, Marty, Joe

American Mineralogist, 104(7), 973–979

Uppermost Triassic phosphorites from Williston Lake, Canada: link to fluctuating euxinic-anoxic conditions before the end-Triassic mass extinction

Larina, Ekaterina, Bottjer, David J, Corsetti, Frank A, Celestian, Aaron J, et al.
Scientific Reports, 9(1), 18790

Pressure-induced hydration and formation of bilayer ice in nacrite, a kaolin-group clay

Hwang, Huijeong, Choi, Jinhyuk, Celestian, Aaron J, Vogt, Thomas, Lee, Yongjae, et al.
ACS Earth and Space Chemistry, 4(2), 183–188

Agate Analysis by Raman, XRF, and Hyperspectral Imaging Spectroscopy for Provenance Determination

Celestian, Aaron, Heginbotham, Arlen, Greenberger, Rebecca, Ehlmann, Bethany, et al.

Mineral preservation and photoprotective attributes of halophilic bacteria and archaea in modern and Permian lakebed environments

Perl, Scott M, Baxter, Bonnie K, Celestian, Aaron J
Studia Universitatis Babeş-Bolyai, Biologia, 64(1)

The Relevance of Hydrothermal Spherules in Astrobiological Exploration of Rocky Planets

Cady, Sherry L, Perl, Scott M, Celestian, Aaron J, et al.
AbSciCon 2019, 321-227

Chemical Biomarker Robustness in Martian Planetary Analogue Evaporite Mineralogy

Perl, Scott M, Celestian, Aaron J, Baxter, BK, Tasoff, Preston
50th Annual Lunar and Planetary Science Conference, 2132, 2930

Hypersaline and Sedimentological Advantages on the Preservation of Extant Biological Activity in Modern Martian Analogue Settings

Perl, SM, Baxter, BK, Celestian, AJ
Mars Extant Life: What's Next?, 2108, 5031

Photobiology and Biogenic Preservation Comparisons Between Pleistocene Evaporite Beds and Buried Permian Brines

Perl, SM, Baxter, BK, Celestian, AJ, Cockell, CS, et al.
The First Billion Years: Habitability, 2134, 1034

Unlocking the Collections: getting comfortable with next-level access to museum minerals

Celestian, Aaron J
Authorea Preprints

Locking away toxic metals into tiny holes

Celestian, Aaron J
Authorea Preprints

The Record of Environmental Change in Northeastern Panthalassa Directly Preceding the End-Triassic Mass Extinction Event

Larina, Ekaterina, Bottjer, David J, Corsetti, Frank A, Celestian, Aaron J, et al.
Geological Society of America Abstracts, 51, 340665

Mineral preservation and photoprotective attributes of halophilic organisms in planetary analogue lake bed environments

Perl, Scott M, Baxter, Bonnie, Celestian, Aaron J, Cockell, Charles S, et al.
AbSciCon 2019, 333-292

2018 – 10 PUBLICATIONS

Preliminary results from Laguna Minucúa: a potentially annually resolved record of climate and environmental change, Oaxaca, Mexico

Goman, Michelle, Joyce, Arthur, Celestian, Aaron J, et al.
Quaternary International, 469, 85–95

Barwoodite, a New Member of the Welinite Group from Granite Mountain, Arkansas

Kampf, Anthony R, Celestian, Aaron J, Nash, Barbara P
The Canadian Mineralogist, 56(5), 799–809

Permian evaporite preservation of chemical biomarkers: Implications for in-situ hydrated mineral and hypersaline brine systems

Perl, Scott, Vaishampayan, Parag, Celestian, Aaron, Baxter, Bonnie, Corsetti, Frank, et al.
42nd COSPAR Scientific Assembly, 42, F3-6

Agate Analysis by Raman, XRF, and Hyperspectral Imaging Spectroscopy for Provenance

Celestian, Aaron, Heginbotham, Arlen, Greenberger, Rebecca, Ehlmann, Bethany, et al.
Gems & Gemology, 54(3)

Investigation of Laminae Formation in Eocene Green River Formation Stromatolites

Mahseredjian, Taleen, Frantz, Carie, Celestian, Aaron J, et al.
AGU Fall Meeting Abstracts, 2018, PP41E-1900

In Situ Aragonite Dissolution in North Pacific: Rate and Role in Modern Ocean Carbon Budget

Dong, Sijia, Rollins, Nick, Celestian, Aaron J, Adkins, Jess F, Berelson, William, et al.
AGU Ocean Sciences Meeting, OC24B-0456

Integrated Optical, Micro-Raman and Compositional Analysis of Rock Samples from Former Talc Mines, Gouverneur Mining District, New York

McNamee, Brittani D, Celestian, Aaron J
Geological Society of America Abstracts, 50, 322067

Genesis of Upper Triassic Phosphatic Ooid Beds from Williston Lake, Canada

Larina, Ekaterina, Bottjer, David J, Corsetti, Frank A, Celestian, Aaron J, Zonneveld, John-Paul
Geological Society of America Abstracts, 50, 321035

A New Study on PIC Dissolution and Sinking Fluxes, Concentrations and Calcite/Aragonite Ratios in the North Pacific

Dong, Sijia, Celestian, Aaron, Turaga, Nitya, et al.
Twenty-eighth Annual Goldschmidt Conference, 591

Evidence for Rapid Rates of Oxygenic Photosynthesis Within Neoproterozoic Stromatolites

Wilmeth, Dylan T, Corsetti, Frank A, Berelson, William M, Celestian, Aaron J, et al.

Geological Society of America Abstracts, 50, 318717

2017 – 3 PUBLICATIONS

Rowleyite, a new mineral with a microporous framework structure

Kampf, Anthony R, Cooper, Mark A, Nash, Barbara P, Cerling, Thure E, Marty, Joe, Celestian, Aaron J, et al.

American Mineralogist, 102(5), 1037–1044

Geobiological Comparisons of Preservation Potential within Hypersaline Mineral-Microbe Systems

Perl, Scott M, Celestian, Aaron J, et al.

AGU Fall Meeting Abstracts, 2017, P41B-2838

Raman and FTIR spectroscopy of methane in olivine

Smith, Annika, Oze, Christopher, Rossman, George R, Celestian, Aaron J

AGU Fall Meeting Abstracts, 2017, V43C-0545

2016 – 6 PUBLICATIONS

Structure determination and time-resolved Raman spectroscopy of yttrium ion exchange into microporous titanosilicate ETS-4

Celestian, Aaron J, Chappell, Caleb J, Rucks, Melinda J, Norris, Pauline

Inorganic Chemistry, 55(21), 11057–11063

Interlaboratory and cross-instrument comparison of Raman spectra of 96 minerals

Dyar, MD, Breitenfeld, LB, Carey, CJ, Celestian, Aaron J, et al.

Lunar Planet. Sci., 46, 2240

In Situ Ion Exchange in Nanoporous Zirconium Silicates

Lively, Jason, Celestian, Aaron, Yan, Bangbo

66th Annual Meeting, MSA

Potential Evidence for Arsenic Mineralization in Mussel Shells in the Upper Green River Basin, Kentucky

Turner, Autumn, Groves, Chris, Celestian, Aaron, Meier, Albert

Mineralogical Crystallography

Celestian, Aaron, Valdez, Nichole

66th Annual Meeting, MSA

Using Raman Spectroscopy to Analyze Oxygen Isotope Ratios in Calcite Speleothems

Koostra, Caleb, Polk, Jason, Celestian, Aaron J, Lively, Jason, et al.

Geological Society of America Abstracts, 48, 281303

2015 – 1 PUBLICATION

Measurements of light-absorbing particles on the glaciers in the Cordillera Blanca, Peru

Schmitt, Carl G, All, John D, Celestian, A, et al.

The Cryosphere, 9(1), 331–340

2014 – 5 PUBLICATIONS

Self-assembled hybrid solids of luminescent Ru(II) polypyridyl complexes and polyoxometalates

Yan, Bangbo, Li, Yanfen, Calhoun, Steven R, Celestian, Aaron J, et al.

Inorganic Chemistry Communications, 43, 23–26

Linking remote and in-situ detection of black carbon on tropical glaciers

Schmitt, Carl G, All, John, Cole, Rebecca J, Celestian, Aaron, et al.

Photogramm Eng Remote Sens, 80(5), 385–390

Measurements of light absorbing particulates on the glaciers in the Cordillera Blanca, Peru

Schmitt, CG, All, JD, Schwarz, JP, Celestian, A, et al.

Cryosphere Discussions, 8(5)

Measurements of Light Absorbing Particles on Tropical South American Glaciers

Schmitt, Carl G, All, John, Celestian, Aaron J, et al.

AGU Fall Meeting Abstracts, 2014, C24A-06

Modeling the Crystal Structure and Ion Exchange Mechanisms in Zorite Using Time-Resolved Raman Spectroscopy

Celestian, Aaron, Chappell, Caleb

Twenty-fourth Annual Goldschmidt Conference, 368

2013 – 5 PUBLICATIONS

In situ Raman spectroscopic study of transient polyhedral distortions during cesium ion exchange into sitinakite

Celestian, Aaron J, Powers, Michael, Rader, Shelby

American Mineralogist, 98(7), 1153–1161

Geochemical and Mineralogical Analysis of Cave Deposits to Reconstruct Climate Change in Barbados

Neeper, Lowell, Polk, Jason, Celestian, Aaron J, et al.

2013 GSA Annual Meeting in Denver

Synthesis and ion exchange studies of porous nano-crystalline materials

Celestian, Aaron

Bulletin of the American Physical Society, 58

Analysis of Fiber Structure in a Concrete Matrix from EBS Coupled to Raman Spectra

Andrew, Keith, Palmquest, Shane, Celestian, Aaron, et al.

Bulletin of the American Physical Society, 58

A Novel Multiproxy Approach to High-resolution Speleothem Paleoclimate Reconstruction in the Caribbean Region

Polk, J, Ouellette, G, Celestian, AJ, et al.

2012 – 7 PUBLICATIONS

Effects of hydration during strontium exchange into nanoporous hydrogen niobium titanium silicate

Kramer, Samantha, Celestian, Aaron J
Inorganic Chemistry, 51(11), 6251–6258

Synthesis and characterization of two- and three-dimensional calcium coordination polymers

Vakiti, Raj Kishore, Garabato, Brady D, Celestian, Aaron, Yan, Bangbo, et al.
Crystal Growth & Design, 12(8), 3937–3943

Identifying Types of Zoning in Minerals Using Raman Microscopy

Flynn, Elaine D, Wulff, AH, Celestian, Aaron J, Rucks, Melinda J
Geological Society of America Abstracts, 44(4), 68

Optical, Raman, and Morphological Characterization of Rock Thin Sections from the Arnold Pit Former Talc Mine

Celestian, Aaron J
Geological Society of America Abstracts, 44(7), 161

New measurements of particulates in glacial snow and ice in the Cordillera Blanca mountains of Peru

All, John, Schmitt, Carl, Celestian, Aaron J, et al.
AGU Fall Meeting Abstracts, 2012, GC11A-0970

Using confocal Raman spectroscopy to analyze nitrate structure at and near the air-water interface

Shockley, Heather C, Nee, Matthew J, Celestian, Aaron
Abstracts of Papers, American Chemical Society, 243

Heavy Metal Sequestration in Nanoporous Minerals

Celestian, Aaron, Powers, Michael, Rader, Shelby
Twenty-second Annual Goldschmidt Conference, 1554

2011 – 1 PUBLICATION

In situ X-ray diffraction study of cesium exchange in synthetic umbite

Fewox, Christopher S, Clearfield, Abraham, Celestian, Aaron J
Inorganic Chemistry, 50(8), 3596–3604

2010 – 6 PUBLICATIONS

Crystal growth and ion exchange in titanium silicates

Celestian, Aaron J, Parise, John B, Clearfield, Abraham
Springer Handbook of Crystal Growth, 1637–1662

In Situ X-Ray Diffraction Study of Cs⁺ Ion Exchange in Umbite

Celestian, Aaron J, Clearfield, Abraham, Fewox, Christopher
2010 GSA Denver Annual Meeting

Cs sequestration into a microporous gallosilicate framework

Leftwich, KM, Celestian, AJ
Geochimica et Cosmochimica Acta, 74(12), A577

The mechanisms of Sr ion exchange into partially niobium substituted sitinakite

Kramer, Samantha, Celestian, Aaron
Twentieth Annual Goldschmidt Conference, 538

Cesium Sequestration into a Microporous Gallosilicate Framework

Leftwich, Kristin, Celestian, Aaron
Twentieth Annual Goldschmidt Conference, 577

The mechanism responsible for extraordinary Cs ion selectivity in crystalline silicotitanate

Celestian, Aaron J, Kubicki, James D, Hanson, Jonathon, Clearfield, Abraham, Parise, John B
Journal of the American Chemical Society, 130(35), 11689–11694

Time-resolved structural characterization of microporous silicotitanates and niobium silicotitanates

Celestian, Aaron Joseph
Dissertation Abstracts International, 69(10)

Strontium ion exchange mechanism in synthetic sitinakite from time resolved studies

Kramer, S, Celestian, AJ
AGU Fall Meeting Abstracts, 2008, V13A-2107

Similarities in 2- and 6-line ferrihydrite based on pair distribution function analysis of X-ray total scattering

Michel, FM, Ehm, L, Liu, G, Han, WQ, Celestian, Aaron J, et al.
Chemistry of Materials, 19(6), 1489–1496

Role of the Hydroxyl-Water Hydrogen-Bond Network in Structural Transitions and Selectivity toward Cesium in Crystalline Silicotitanate

Celestian, Aaron J, Parise, John B, Smith, Ronald I, Toby, Brian H, Clearfield, Abraham
Inorganic Chemistry, 46(4), 1081–1089

The origin of ion exchange selectivity in a porous framework titanium silicate

Celestian, Aaron J, Clearfield, Abraham
Journal of Materials Chemistry, 17(46), 4839–4842

Dehydration-induced water disordering in a synthetic potassium gallosilicate natrolite

Lee, Yongjae, Kim, Sun Jin, Bull, Ivor, Celestian, Aaron J, Parise, John B, et al.
Journal of the American Chemical Society, 129(44), 13744–13748

Technical Report: Growth of Environmental Science at the NSLS

Northrup, Paul, Lanzirotti, Antonio, Celestian, Aaron

Synchrotron Radiation News, 20(3), 6–13

2006 – 5 PUBLICATIONS

Solid-state structural characterization of a rigid framework of lacunary heteropolyniobates

Nyman, May, Celestian, Aaron J, Parise, John B, Holland, Gregory P, Alam, Todd M

Inorganic Chemistry, 45(3), 1043–1052

In situ type study of hydrothermally prepared titanates and silicotitanates

Clearfield, A, Tripathi, A, Medvedev, D, Celestian, AJ, Parise, JB

Journal of Materials Science, 41(5), 1325–1333

Temperature stability of a molecular sieve crystalline silicotitanate with the mineral sitinakite topology

Galloway, Melissa M, Celestian, Aaron J, Parise, John B, Clearfield, Abraham, Perry, Houston P

Abstracts of Papers, American Chemical Society, 231

Influence of hydroxyl-water interaction on structural transformations during Cs+ exchange in microporous H-CST

Celestian, AJ, Parise, JB, Narayan, J, Kubicki, JD, Clearfield, A

Geochimica et Cosmochimica Acta, 70(18), A91

GEOC 133 — Size and structure of ferrihydrite from PDF analysis

Michel, F Marc, Antao, Sytle M, Celestian, Aaron J, Schoonen, Martin A, et al.

Abstracts of Papers, American Chemical Society, 232

2005 – 7 PUBLICATIONS

High-pressure neutron diffraction study of superhydrated natrolite

Colligan, Marek, Lee, Yongjae, Vogt, Thomas, Celestian, Aaron J, Parise, John B, et al.
The Journal of Physical Chemistry B, 109(39), 18223–18225

Optimizing synthesis of $\text{Na}_2\text{Ti}_2\text{SiO}_7 \cdot 2\text{H}_2\text{O}$ and ion exchange pathways for Cs-CST determined from in situ synchrotron X-ray powder diffraction

Celestian, AJ, Medvedev, DG, Tripathi, A, Parise, JB, Clearfield, A
Nuclear Instruments and Methods in Physics Research B, 238(1–4), 61–69

Tribasic lead maleate and lead maleate: Synthesis and structural and spectroscopic characterizations

Bonhomme, François, Alam, Todd M, Celestian, Aaron J, Tallant, David R, Nyman, May, et al.
Inorganic Chemistry, 44(21), 7394–7402

In situ neutron and X-ray scattering studies of Cs⁺ ion exchange sequestration into the crystalline silicotitanate (CST) molecular sieve

Celestian, AJ, Parise, JB, Smith, RI, Park, HS, Clearfield, A
Abstracts of Papers, American Chemical Society, 230

2004 – 5 PUBLICATIONS

Time-Resolved Diffraction Studies of Ion Exchange: K⁺ and Na⁺ Exchange into (Al, Ge) Gismondine

Celestian, Aaron J, Parise, John B, Goodell, Carmen, Tripathi, Akhilesh, Hanson, Jonathan
Chemistry of Materials, 16(11), 2244–2254

Crystallization of sodium titanium silicate with sitinakite topology: Evolution from the sodium nonatitanate phase

Medvedev, Dmitri G, Tripathi, Akhilesh, Clearfield, Abraham, Celestian, Aaron J, Parise, John B, Hanson, Jonathan
Chemistry of Materials, 16(19), 3659–3666

In situ X-ray diffraction study of Cs ion exchange into H-TS

Celestian, Aaron J, Medvedev, Dmitri G, Tripathi, Akhilesh, Parise, John B, Clearfield, Abraham
2004 Denver Annual Meeting, GSA

2003 – 3 PUBLICATIONS

Phosphor imaging as a tool for in situ mapping of ppm levels of uranium and thorium in rocks and minerals

Cole, Jennifer M, Nienstedt, Jessica, Spataro, Glenn, Celestian, Aaron J, et al.

Chemical Geology, 193(1–2), 127–136

(K₄Li₄)Al₈Ge₈O₃₂·8H₂O: an Li⁺-exchanged potassium aluminogermanate with the zeolite gismondine (GIS) topology

Celestian, Aaron J, Parise, John B, Tripathi, Akhilesh, Kwick, Åke, Vaughan, Gavin MB

Crystal Structure Communications, 59(8), i74–i76

In situ synchrotron X-ray structural studies of crystal growth and ion exchange in tunnel-type titanosilicates

Tripathi, A, Medvedev, D, Clearfield, A, Celestian, AJ, Parise, JB, Nyman, M, Hanson, J

Abstracts of Papers, American Chemical Society, 225

2002 – 3 PUBLICATIONS

A new polymorph of eucryptite (LiAlSiO₄), ε-eucryptite, and thermal expansion of α- and ε-eucryptite at high pressure

Zhang, Jianzhong, Celestian, Aaron, Parise, John B, Xu, Hongwu, Heaney, Peter J

American Mineralogist, 87(4), 566–571

Ion Exchange of Gismondine

Celestian, Aaron Joseph

M.S. Thesis, Stony Brook University

Instability of Beta-Eucryptite under Pressure and Thermal Expansion of ε-LiAlSiO₄ at 1.9 GPa

Zhang, JN, Celestian, A, Parise, J, Xu, H, Heaney, P

American Mineralogist, 877 (BNL-74038-2005-JA)

Phase Transformations in the GIS Structure upon Ion Exchange

Parise, J, Goodell, C, Celestian, A, Tripathi, A
AGU Fall Meeting Abstracts, 2001, V32A-0956

MEDIA & OUTREACH

BLOG & PODCAST

Pocketful of Xtals — Host & Creator (Substack, Spotify, Apple Podcasts, iHeart)

TELEVISION

The UnXplained — History Channel (TV Series, 2020–2023, 3 episodes: S1E14 "Mysterious Stones," S5E3 "Curses Unleashed," S5E7 "Sacred Rituals")

Expedition Unknown: Legend of the Crystal Skull — Discovery Channel (2019)

Spectrum News 1 — Lead contamination expert

Univision — Interview

FILM & MUSEUM VIDEO

The Color of Ink — Consulting Geologist (2022)

Behind the Scenes with Mineral Sciences — NHMLAC (2017)

Diamonds: Rare Brilliance Exhibition Walkthrough — NHMLAC, with Elena Manferdini (2017)

RADIO

NPR TakeTwo

PRINT MEDIA

Eos (AGU Earth and Space Science News) — Featured scientist analyzing ash samples

from 2025 California wildfires

New York Times — Parenting Section: The Trouble With Amber Teething Necklaces

National Geographic — Rarest Form of Gold

Reuters — Mystery black diamonds 'The Enigma' (also BBC, NASDAQ, Livemint, Yahoo, Taipei Times)

LA Times — Plants section

Prospering Backyards — Cover Story

MINERAL & GEM SOCIETIES

Tucson Fine Mineral Gallery — Tucson Gem & Fine Mineral Show, February 4, 2026: "Beyond the Display Case: How Modern Mineralogy is Fighting Disease"

Tucson Fine Mineral Gallery — Tucson Gem & Fine Mineral Show, February 2025

Mineral Talks LIVE — Episode 66, October 2022

MSSC — Zoom, April 9, 2021: "The Lithium Gold Rush"

PROFESSIONAL SYMPOSIA

Mineralogical Society of America Centennial Symposium — "Unlocking the Collections: Making Minerals Accessible in the Next 100 Years" (with Kim Tait, Royal Ontario Museum)

Sinkankas Symposium, GIA — "Using Gems to Illuminate Art and Science"

CIVIC & COMMUNITY

Rotary Club of Beverly Hills — Beverly Hills Hotel, March 23, 2026

Adventurers' Club of Los Angeles — February 29, 2024: "Searching for Life Deep Underground"

Dahlia Heights Elementary — Career Day keynote

ART & INTERDISCIPLINARY

NAHR (Nature, Art & Habitat Residency) — Multiple talks, Taleggio Valley, Italy / Santa Ynez, CA; minerals-themed programming 2025–2026

MUSEUM PROGRAMS (NHMLAC)

Meet the Scientist — Video series

Happy Hour with Dr. Celestian

Members Q&A — February 11, 2021 (Zoom)

Mineral Talks LIVE

PANELS & INVITED TALKS

First Fridays Panel — NHMLAC: The Search for Life at the Cosmic Scale

Family Office Panel — Miami Beach: The Future of Regenerative Energy (Invited)

SMMP Panel — "The Future of Mineralogy Education," February 12, 2026

TECHNICAL SKILLS

ANALYTICAL TECHNIQUES

X-ray Diffraction (XRD)

Raman Spectroscopy

SEM/TEM Imaging

Near-Infrared Imaging

SOFTWARE & PROGRAMMING

Python — Data Analysis

RamanLab — Developer

Crystallography Software

Machine Learning

PROFESSIONAL AFFILIATIONS

Mineralogical Society of America
